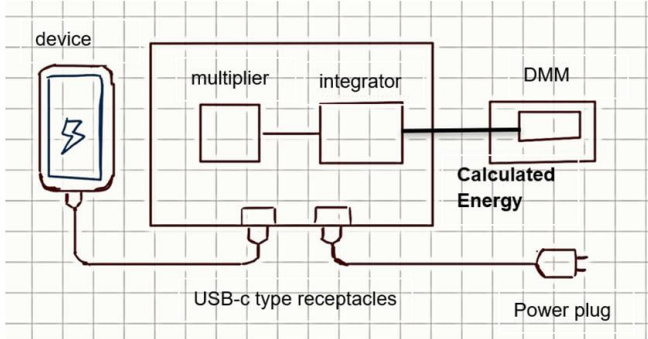
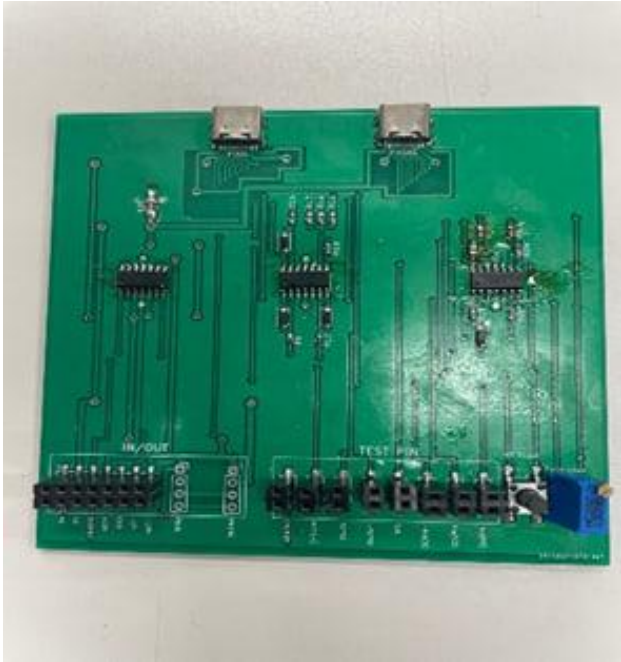


Project: Electric Energy Calculator using Op-Amp Integrator	
Location:	Seoul National University (2nd Year, Spring Semester)
Duration:	April 2024 – June 2024
Project Overview	
Description:	Development of an analog energy calculator utilizing operational amplifiers.  Figure 1. Design of the Energy Integrator
Role & Responsibilities:	• Concept Development: Proposed the core project concept and led the initial design phase. • Circuit Implementation: Constructed the integrator circuit and verified theoretical assumptions. • PCB Design: Collaborated on the schematic and PCB layout design using KiCad.
Technical Contributions	The project lifecycle encompassed conceptualization, circuit design, physical implementation, and final manufacturing. • Design Phase: I spearheaded the project initiation by proposing the core concept. With the help of my teammates, I proposed the idea of carrying out energy calculation on a circuit using power delivered to devices when charging . • Implementation & Verification: I was responsible for validating our initial

	assumptions by prototyping the integrator circuit. I subsequently integrated this module with the multiplier circuit to complete the system logic. PCB Layout: Following circuit verification, I assisted in translating our schematics into a PCB layout using KiCad, preparing the final design files for the manufacturer.  <i>Figure 2. Final look of our PCB board</i>
Key Takeaways & Professional Growth	This project provided enduring lessons in engineering collaboration and project management. 1. Root Cause Analysis and Problem Solving: Working with talented teammates taught me the value of a solution-oriented mindset. When actual results deviated from our simulations, rather than seeking immediate external help, we focused on rigorous troubleshooting. I learned to systematically trace errors back to their source—often discovering simple oversight or calculation errors. This experience taught me the importance of acknowledging mistakes early and performing a "root cause analysis" to resolve technical failures efficiently.

	2. Project Management: Macro vs. Micro Perspective: I developed the ability to balance detailed execution with the "big picture." Initially, I had a tendency to fixate on the minute details of individual components, which slowed progress. My teammates helped me shift focus toward establishing the general system architecture first and refining details later. This agile approach prevented circular problems and ensured we met our reporting milestones. I now approach engineering problems by adhering to the main objective while managing stages of varying detail levels. A photograph of three young men standing in a laboratory or workshop. The man on the left is wearing a dark button-down shirt and light-colored pants, making a peace sign. The man in the middle is wearing a white t-shirt and light-colored pants, also making a peace sign. The man on the right is wearing a green polo shirt and dark pants, making a peace sign. They are standing in front of a table with electronic equipment, including a laptop and various cables. In the background, there are shelves with equipment and a sign that reads "I hear and I forget, I see and I understand". <i>Figure 3. Final day of presentation</i>
Skills	
Technical & Hardware	• Circuit Design: Analog Circuit Implementation, Operational Amplifiers (Op-Amps), Integrators, and Multiplier Circuits . • PCB Engineering: Schematic Design, PCB Layout, and Manufacturing Preparation using KiCad .

		• Verification: Hardware Prototyping, Circuit Testing, and Validation .
Professional Methodological	&	• Problem Solving: Root Cause Analysis, Error Tracing, and Debugging . • Project Management: Strategic Planning (Macro-level execution), Progress Reporting, and Time Management . • Communication: Technical Presentation and Cross-functional Team Collaboration .